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AI Based Autonomous Control for Oil and Gas Using FKDPP Technology

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Abstract

This paper demonstrates the application of the Autonomous Reinforcement learning based AI Controller, based on the Factorial Kernal Dynamic Policy Programming (FKDPP) algorithm in an upstream oil and gas processing facility to enhance process stability and reduce manual operator interventions. FKDPP utilizes reinforcement learning to address operational challenges, such as liquid level and pressure fluctuations in separators caused by unstable fluids and uneven crude distribution. The AI controller analyzes inputs from multiple separators simultaneously, managing final control elements to optimize conditions with minimal reliance on traditional PID control. Trained on dynamic simulators and plant historical data, it adjusts separator inlet level control valves (LCVs) to stabilize levels, gas flow, suction pressure, and inlet pressure. Integrated via OPC communication at Level 3 of the control system, it considers upstream and downstream dynamics to mitigate disturbances. Currently operating in closed loops on 3 of 5 separators, initial performance shows promising potential for scaling across platforms, reducing intervention, and improving operational efficiency. Future phases aim to achieve full autonomous control on all 5 separators.

Introduction

Well Fluid from remote Well Head Towers through the 10 incoming radials is routed to a 16" oil manifold (known as Ring Main) located on Separators platform. The ring main supplies the crude oil to the individual five separators (328-C-102~106). The function of the 2- phase separator is to partially degas the crude oil prior to transfer to Das Island refinery. Each separator is designed to handle well fluid equivalent to 60 MSTBPD oil production and operates at 450 psig (max). Flashed/separated gas from all the five separators are combined to a common header and routed to the gas scrubbers located on the pump platform for removal of entrained liquid (oil carry ov er), if any. Separated liquid (oil + water) from each running separator flows into a common 20" header. Booster pumps provide necessary NPSH to MOL Pumps (2 operating + 1 Standby). Mol Pumps further pump crude to the required pressure based on MOL pipeline back/operating pressure. Each separator is provided with level indicator and controller regulating the LCV on the discharge of respective booster pump. To minimize the impact of mal distribution of feed to separator and manage equal flow distribution and balancing between separators, LCVs at inlet to each separator is provided as shown in Figure 1. The Inlet controller controls the inlet feed to each separator and the outlet LCV controller maintains the minimum level in each separator and prevents the separator from the low-level trip.

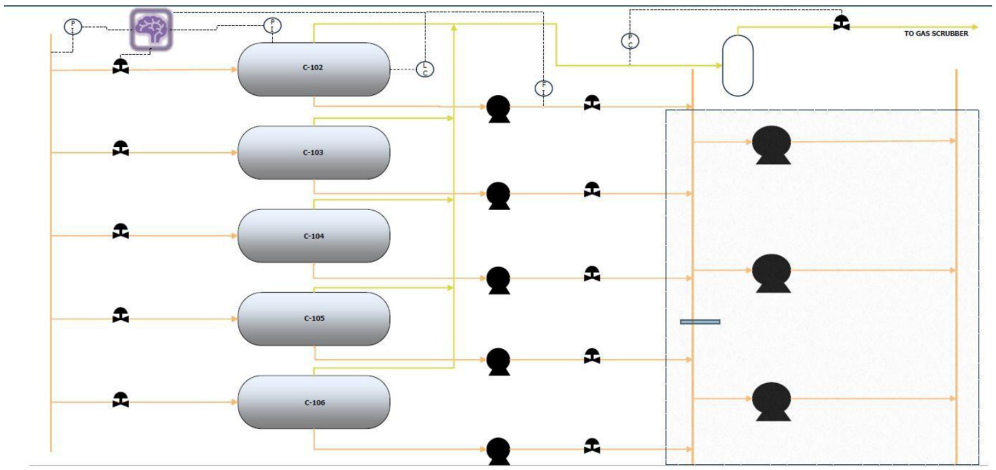

Figure 1—Overview: AI Controlled Oil Separators

Background and Challenges

The complex oil separation system consists of five two-phase separators. While PID control systems have been implemented, operational level management is largely handled manually by on-site operators, leading to the following issues:

1. Occurrence of liquid carryover and challenges in maintaining consistent oil/gas separation quality.
2. Overshooting resulting in unnecessary flaring.
3. Increased labor costs due to operator intervention.

Significant fluctuations in the properties of production fluids from the wellheads, coupled with the very short residence time of fluids within the oil ring main and the uneven distances between the entry points of production fluids into the oil ring main and each separator, result in uneven distribution of fluids across the five oil separators. Moreover, the variation in pipe lengths significantly exacerbates the uneven fluid distribution under high GOR (Gas-to-Oil Ratio) conditions, further complicating the effective operation of the separation system.

Uneven fluid distribution into the separators leads to a rapid influx of gas into C-102 & C-106, which is located closest to the entry point of production fluids into the oil ring main. In contrast, separators positioned further downstream (C-103, C-104, and C-105) receive minimal amounts of gas from the oil ring main. Although the maximum allowable overhead gas flow rate for the separators is 75 MMSCFD, instances have been observed where the gas flow rate consistently exceeds 90 MMSCFD, presenting operational challenges and risks. Under high GOR conditions, a soft clamp had been implemented in the current operation of the inlet level control PID loops to limit the maximum opening of the inlet control valves based on operating conditions, in order to prevent excessive opening of the separator inlet controls. However, despite the use of soft clamps to restrict the valve openings, overhead gas flow rates from separators C-102 and C-103 frequently exceed the allowable threshold of 75 MMSCFD, with values often observed as high as 93 MMSCFD. This results in increase gas carry over from the separators and starvation of gas to C-103, C-104 and C-105.

Methodology and Execution Approach

In this pilot study, an AI model was developed using reinforcement learning algorithms which monitors parameters such as separator pressure, separator liquid level, downstream flow rate, overhead gas flow rate, upstream oil ring main pressure and MOL pump suction pressure. The study focused on implementing the constructed AI controller to regulate the opening of the separator inlet control valves and evaluate if A.I based technology could be successfully and safely deployed in a production facility for real time control to over come the challenges faced.

Expected Benefits of AI Autonomous Control Implementation in Oil Separators

1. Effective Separator Liquid Level Control
   Traditional PID controllers regulate liquid levels based solely on level fluctuations. However, the liquid level in separators is influenced by multiple factors, such as oil ring main pressure, header pressure, and MOL pump speed. FKDPP takes these dynamic process conditions into account, enabling autonomous control to stably maintain the liquid level at approximately 40% and prevent high gas flow rates resulting in liquid carryover.
2. Prevention of Liquid Carryover
   Under high GOR conditions, FKDPP adjusts the inflow rate to the separators to limit the gas overhead flow to less than 80 MMSCFD, thereby mitigating liquid carryover. As a result, appropriate gas flow distribution among all separators is achieved.
3. Reduction in Operator Intervention
   Currently, a soft clamp is applied to the inlet control valve to limit its maximum opening in order to prevent excessive gas flow from separators. However, due to dynamic operating conditions, it is challenging to generalize the maximum soft clamp setting, requiring frequent manual adjustments by operators based on specific operating conditions. The FKDPP effectively controls the inlet control valve to maintain a stable level during normal operations and prevent carryover during upset scenarios. This eliminates the need for soft clamp settings on the valve's maximum opening, significantly reducing operator intervention.

AI Model Development

The AI model was developed using historical data provided by ADNOC Offshore, covering the period from January 2024 to April 2025. Development was conducted utilizing a dynamic simulator model located at manufacturer's office. Initially, the model was developed specifically for the C102 one separator, and subsequently, it was extended to other separators in a scalable manner.

The above describes an overview of reinforcement learning-based control models from the perspective of autonomous plant operation. In such models, the agent learns through iterative interactions with the environment. During each iteration, the agent selects an action (e.g., adjusting valve positions) based on the current state (e.g., process parameters). Subsequently, the agent uses the res ulting reward (e.g., the degree of achievement of control objectives) to refine its policy, thereby improving decision-making over successive iterations. Factorial Kernel Dynamic Policy Programming (FKDPP) is a reinforcement learning algorithm designed for such applications. It is distinguished by its use of kernel functions to enable smooth policy updates, ensuring iterative improvement of the decision-making framework while maintaining stability in dynamic and complex systems.

Figure 3 illustrates the control model for the C102 separator. In this model, disturbance variables, including inlet pressure (I/L Pressure), header pressure, MOL pump pressure, header backline pressure (Header BL Pressure), and MOL pump discharge backline pressure (Mol Pump Dis BL Pressure), are utilized as components of the reward function. Status parameters are defined as the state space of the system.

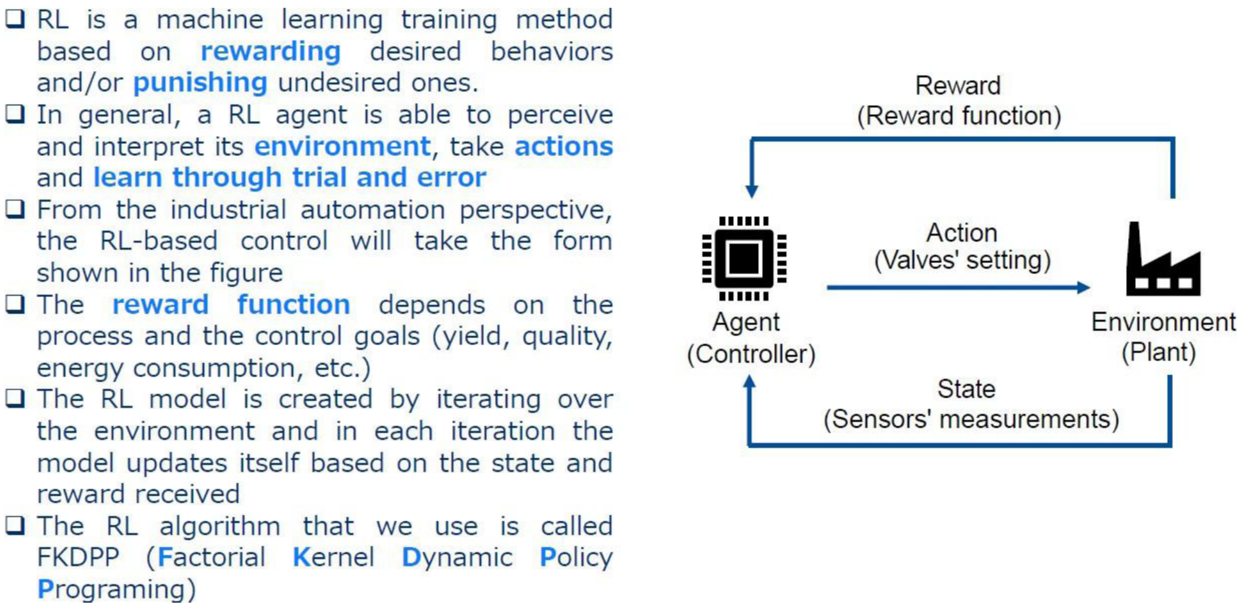

Figure 2—Overview: Reinforcement Learning for Autonomous Plant Control

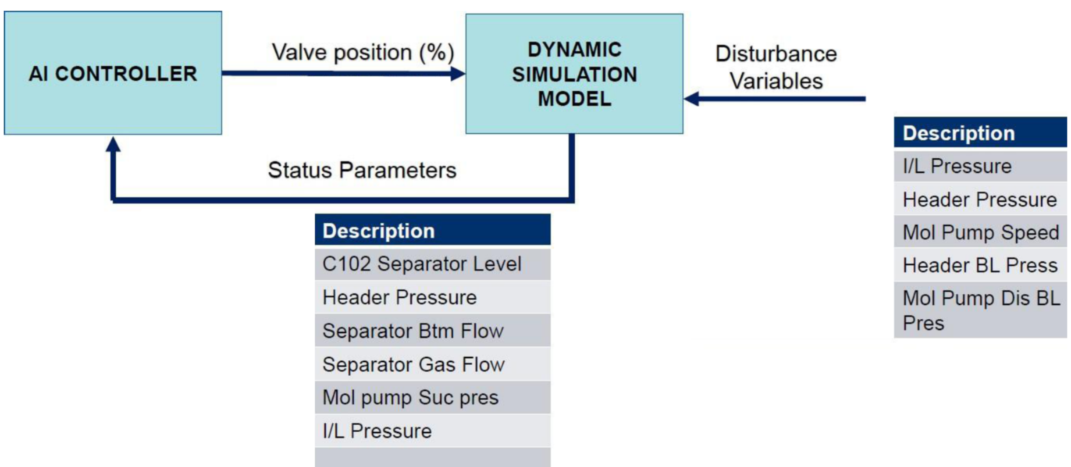

Figure 3—AI Control Model of the C102 Separator

The AI controller selects actions, such as valve adjustments, based on the system's current state, aiming to maximize the reward. The learning process is structured as an iterative state-action-reward cycle, allowing the controller to continuously refine and optimize its policy through repeated dynamic interactions with the environment.

AI-driven control is activated only when the system operates within predefined acceptable boundaries. If the system deviates from these boundaries and remains outside this envelope for a specified duration, the AI control is automatically disengaged, and the system seamlessly reverts to conventional PID control. This safety mechanism ensures reliable operation during abnormal process conditions and bump less switchover between AI and PID MODES.

Commissioning

Open-loop Commissioning

The Open Loop Commissioning was conducted in April 2025, prior to the Closed Loop Commissioning. Its primary objective was to set up the AI Controller and establish an environment enabling monitoring and evaluation of its control behavior. This preparatory phase ensured that the system was appropriately configured for subsequent closed-loop operation under real-time conditions.

The Open Loop Commissioning involved the following activities:

1. Setup of the AI Controller
   - Installation of the AI software and configuration of the operational environment within the plant to enable the AI Controller.
2. Modification of Existing DCS Logic and Graphics
   - Adjustments to DCS (Distributed Control System) logic and graphics were conducted to prepare for AI-based control operations.
3. Historical Tag Registration in ExaQuantum
   - Tags for AI Controller inputs and outputs were saved in ExaQuantum, the process data collection system. This allowed operational data to be archived and made accessible for subsequent analysis.
4. Establishment of Communication Between the AI Application and Relevant Systems
   - Robust communication channels were established between the AI application and the following systems:
     - Human Interface Station
     - Process Data Historization System
     - OPC Communication System

While the Open Loop Commissioning phase did not involve actual control by the AI Controller, AI output data was recorded in Process Data Historization System for historization. The recorded performance data of the AI Controller was analyzed over several weeks under dynamic plant conditions. Based on this analysis, fine-tuning of the AI Controller was conducted, and the AI model was updated. These adjustments provided an optimized configuration for proceeding to the Closed Loop Commissioning phase.

Closed-loop Commissioning

During A.I. Close loop commissioning, an updated A.I. control model is placed on the A.I. server. The A.I. controller output is engaged to the final control element. After the A.I. Close loop commissioning, the performance of the A.I. controller to the plant dynamics is monitored. The activation of AI control was conducted in collaboration with on-site supervisors and operators. To ensure a controlled and risk-averse implementation, the activation sequence followed the order of systems with less stringent operational conditions: C104 → C103 → C105 → C102, at a pace of one system per day. It should be noted that C106 was excluded from this activation process as it was offline due to ongoing maintenance activities.

A.I. Control Performance

The A.I. control is enabled in a sequence of separators with low gas flowrate to high gas flowrate so that the excess gas from separators with high gas flowrate can be distributed to separators with low gas flowrate. Separator C103 & C104 are identified as the low gas flowrate and separator C102 & C105 are identified as high gas flowrate.

The Separator C104 is selected to witness the A.I. control disengage and P-ID setpoint ramp-up/down. A.I. mode is enabled in the Separator C104 and disabled manually, during which the auto ramp-down functionality of P-ID setpoint was observed. Manual intervention to adjust the PID controller output was also tested. Upon disabling the A.I. mode, both an alarm message and sound were triggered. It has been confirmed that the current logic is functioning as expected.

A.I. Control - Separator C104

The A.I. mode was enabled for separator C104, and the A.I. controller's output responded correctly to level and gas flow rate variations despite upstream/downstream disturbances. However, after approximately 15 minutes, the A.I. controller automatically disengaged due to safety logic related to the MOL pump suction pressure upper threshold of 545 psig. An alarm message indicating A.I. disabl e was observed. Subsequently, the upper threshold of MOL pump suction pressure was adjusted to 560 psig for all separators, and the A.I. mode for separator C104 was re-enabled. During this phase, no auto disengagement of the A.I. controller occurred, and the controller remained in A.I. Enable mode for an extended duration.

It was decided to continue running the A.I. controller, while the A.I. mode for other separators was scheduled for activation at a later time. It was reported from operations that the auto disengagement of separator C104's A.I. controller occurred later due to sudden dips in MOL pump suction pressure (PIC201.PV) below 510 psig at different intervals. However, operations re-enabled the A.I. mode and observed that the A.I. controller could effectively maintain the separator level around 40% despite dynamic process conditions.

The A.I. controller output for separator C104 was directionally correct across both higher and lower inlet flow rates. The level was consistently maintained around 40%, with a deviation of approximately ±2.5–3%. The operations team suggested increasing the A.I. controller's response rate to minimize the deviation between the set value (SV) and the process value (PV).

Whenever A.I. mode disengagement occurs, the transition from A.I. control to PID control is seamless and smooth. Similarly, the transition from PID control to A.I. control is seamless, ensuring operational stability throughout the process.

A.I. Control - Separator C103

Based on observations from separator C104, the cycle time of separator C103 was increased from 4 seconds to 8 seconds to improve the response rate of the inlet control valve compared to separator C104. Initially, the cycle time of separator C104 was left unchanged to allow comparison of the A.I. control performance between separator C103 (with a cycle time of 8 seconds) and separator C104.

The A.I. control was enabled for separator C103, and the A.I. controller effectively adjusted the inlet level control valve to maintain level and gas flow rate within operational limits, despite dynamic upstream and downstream conditions. Observations comparing the

A.I. control response to level variations between separator C104 (with updated model parameters) and separator C103 (with existing model parameters) indicated satisfactory performance over the evaluation period. Based on this performance, it was decided to implement the same model parameters used in separator C103 for separator C104.

Subsequently, A.I. control for separator C104 was manually disabled to update tuning settings to match those of separator C103, which remained active in A.I. mode during the update. An alarm notification was observed when the A.I. controller for separator C104 was disabled. Following the update, A.I. control for separator C104 was re-enabled, and both separators, C103 and C104, continued operating in A.I. control mode without disengagement.

It was decided to maintain A.I. control mode for separators C103 and C104, and preparations were made to enable A.I. control for separator C105 as the next step.

The following are the model parameters set in separators C103 and C104:

- Control Cycle – 8 seconds
- Level Low Limit @ High Gas Flowrate – 35 %

- High Gas Flowrate Limit – 77 MMSCFD

The separator C103 & separator C104 are in A.I. mode and the level in the separators is maintained around 40% with a deviation of approx. +/- 2% in normal conditions. Disturbance from downstream pressure around 50 psig has also been observed, still A.I. controllers could maintain the separators C103 & C104 level around 40% and limit the gas flow rate less than 80 MMSCFD.

A.I. Control - Separator C105

The A.I. controller was enabled for separator C105 using the same parameters as separator C103. It was observed that the operation of separator C105 is more sensitive to upstream gas/liquid pockets and downstream pressure fluctuations compared to separators C103 and C104. Despite these variations, the A.I. controller consistently adjusted the valve position correctly in all scenarios.

Several instances of automatic mode switching from A.I. control to PID control occurred when the liquid level in separator C105 reached the safety logic lower limit. Each time, the A.I. mode was re-enabled after adjusting model parameters to achieve optimal control performance. Separators C103 and C104 remained in A.I. mode with stable operation throughout the observation period. During extended operation, the controller for separator C105 switched to PID mode when the safety logic triggered due to a low liquid level.

Further adjustments were made to enhance the controller's response rate by increasing the valve step size (action parameter) while maintaining the control cycle time. Additionally, the high gas flow rate threshold was revised to improve performance under dynamic upstream conditions. During high gas flow scenarios, a sudden influx of gas-liquid pockets was observed to cause rapid increases in liquid levels, necessitating manual disengagement of A.I. mode to prevent potential liquid carryover.

Subsequent updates included refining the low-level limit during high gas flow conditions and reducing the control cycle time for improved responsiveness. Separator C105 has since operated stably in A.I. mode without further disengagement.

The following are the updated parameters set for separator C105:

- Control Cycle – 2 seconds
- Level Low Limit @ High Gas Flowrate – 38%
- High Gas Flowrate Limit – 80 MMSCFD
- Outlet PID controller set point – 38%

It was observed that the liquid level occasionally dropped to around 31% when downstream pressure rapidly decreased and then recovered. To prevent A.I. disengagement due to the safety low-level limit, the threshold for the separator liquid level was lowered to 29%, while the emergency shutdown (ESD) low trip remained at 20% across all separators. Separators C103 and C104 continued stable operation in A.I. control mode, whereas separator C105 occasionally experienced disengagement when the liquid level reached the revised 29% limit.

Following updates to the model parameters, separator C105 demonstrated improved resilience to low-level scenarios. However, high liquid levels were occasionally observed during sudden liquid inflow events from upstream. Feedback from the operations team indicated that performance and response rate for separator C105 had improved significantly with this update.

Separators continued to meet production rate targets after optimizations. At one point, the A.I. controller for separator C105 was manually disengaged temporarily to prioritize achieving monthly production goals during periods of operational adjustments.

A.I. Control - Separator C102

Separator C102 underwent A.I. control testing using the same model parameters as separator C105. Automatic mode switching from A.I. control to PID control occurred when the liquid level reached the safety low limit of 29%. Each time, the A.I. mode was re-enabled after adjustments to the model parameters to optimize control performance.

The following are the parameters set for Separator C102:Control Cycle – 8 seconds

- Level Low Limit @ High Gas Flowrate – 35 %
- High Gas Flowrate Limit – 77 MMSCFD
- Outlet PID controller set point – 38%

While operating in A.I. mode, Separator C102 effectively distributed gas to Separators C103 and C104, maintaining balanced flow distribution across the units. Separators C102, C103, and C104 remained in A.I. mode until 1:00 PM. At that time, Separator C102's

A.I. control auto disengaged after the liquid level dropped to the safety low limit of 29%, triggered by upstream choke adjustments that introduced gas pockets.

Toward Improving and Expanding AI Control Performance

Adjustments to the PID controller tuning of the outlet level controllers for separators C102 and C105 were requested to synch ronize operations between the inlet and outlet control valves, improving liquid level and gas flow rate management. Currently, the outlet controller on C102 responds too quickly due to its existing PID tuning parameters ($P \approx 70$, $I \approx 45$). It was recommended to maintain Separator C102 in PID mode for the time being.

Separators C103, C104, and C105 will continue operating in A.I. control mode until Separator C106 is returned to service. Once C106 is back online, its A.I. mode will be activated, utilizing similar model parameters as those for Separator C105. Remote support will be provided during this transition.

The A.I. controller performance for all separators will be monitored for the evaluation period. After the evaluation period, A.I. control for Separator C102 will be activated. It was advised to configure the outlet control valve of Separator C102 to operate in A. I. mode rather than modifying the existing PID tuning parameters.

Conclusion

This paper presents the pioneering application of AI-based autonomous control using the Factorial Kernel Dynamic Policy Programming (FKDPP) algorithm in an oil and gas offshore processing facility. This represents the world's first implementation of FKDPP technology in such a complex industrial setting, showcasing its ability to manage dynamic process conditions and enhance operational efficiency. By leveraging reinforcement learning, the AI controller demonstrated effective control of critical parameters, such as separator liquid levels and gas flow rates, under varying upstream and downstream disturbances. The autonomous system successfully reduced operator interventions, minimized safety risks due to liquid carryover, and maintained process stability while integrating seamlessly with conventional PID control for safety transitions. Continuous optimization of model parameters ensured robust control performance across multiple separators, providing scalable insights for future implementations.

Although the five separators were designed to be identical, variations in physical construction, piping isometrics, and resulting gas- liquid dynamics led to distinct operational behaviors for each unit. Leveraging the FKDPP algorithm, we successfully developed a single adaptive AI model capable of accommodating these behavioral differences and operational constraints—eliminating the need for maintaining multiple specialized models. This marks a significant advancement in AI-driven systems, challenging the conventional belief that AI solutions must be tightly tailored to narrowly defined datasets. Our approach demonstrates that with the right architecture, a unified model can robustly generalize across complex, real-world variations and uncertainties. Moreover, the introduction of FKDPP is anticipated to contribute economically, as stabilizing separation processes not only reduces operational inefficiencies but also has the potential to increase production rates over time. This transformative technology paves the way for broader industrial applications, offering significant advantages in achieving higher production targets while maintaining safe and sustainable operations.

References

Cui, Y., Zhu, L., Fujisaki, M., Kanokogi, H., & Matsubara, T. (2018). Factorial Kernel Dynamic Policy Programming for Vinyl Acetate Monomer Plant Model Control. Proceedings of the IEEE International Conference on Automation Science and Engineering (CASE).

Park, J. (2025). Reinforcement Learning for Process Control: Review and Perspectives. *Processes*, **13**(6), 1791.

Yokogawa Electric Corporation. (2024). *FKDPP AI Controller System Architecture. Technical White Paper*.

Sutton, R. S., & Barto, A. G. (2018). *Reinforcement Learning: An Introduction* (2nd ed.). MIT Press.